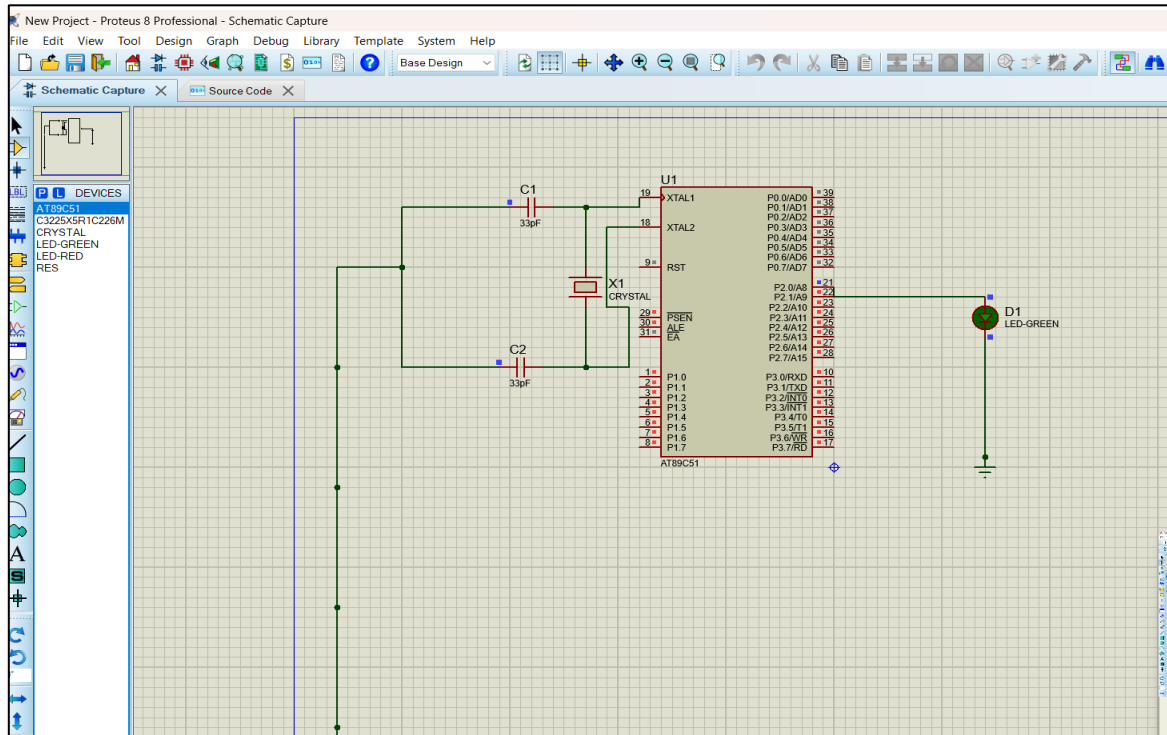


ECA1411-EMBEDDED SYSTEM

LAB MANUAL

Exp No: 1 FLASHING OF LED USING AT89C51 MICROCONTROLLER USING PROTEUS



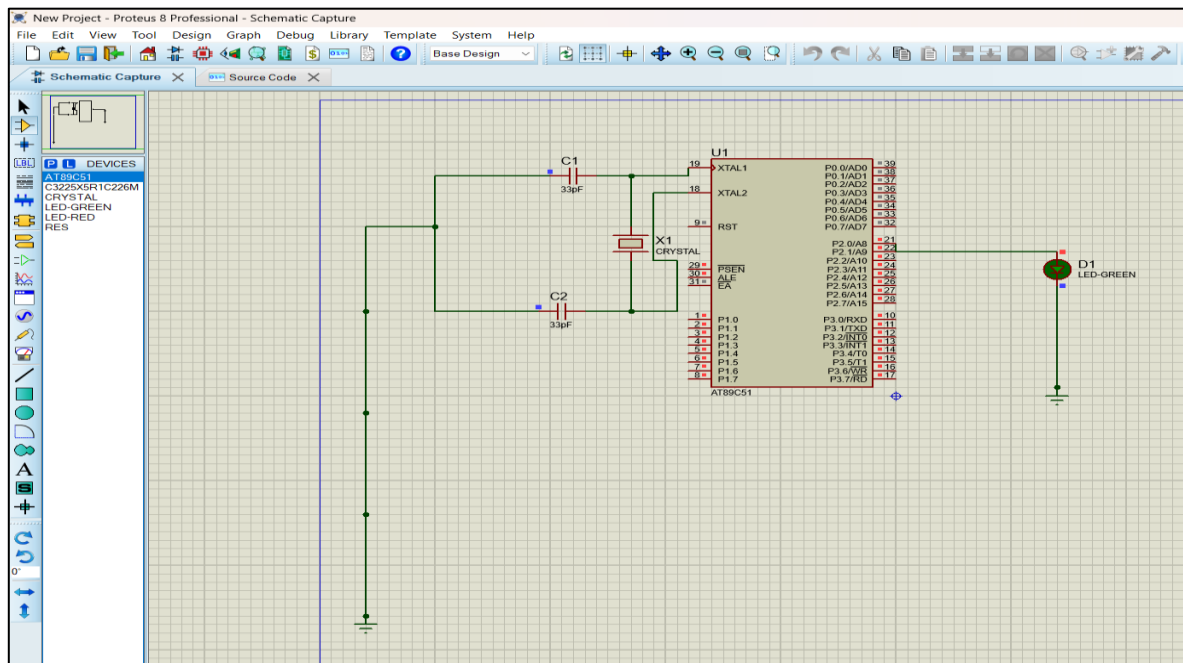
SOURCE CODE:

```
New Project - Proteus 8 Professional - Source Code
File Project Build Edit Debug System Help
Schematic Capture Source Code
8051 CPU Source Code - U1
Debug(Debug.SDI)

-----ORG 0000H
0000 UP:
0000 SETB P2.0
0002 ACALL DELAY
0004 CLR P2.0
0006 ACALL DELAY
0008 SJMP UP

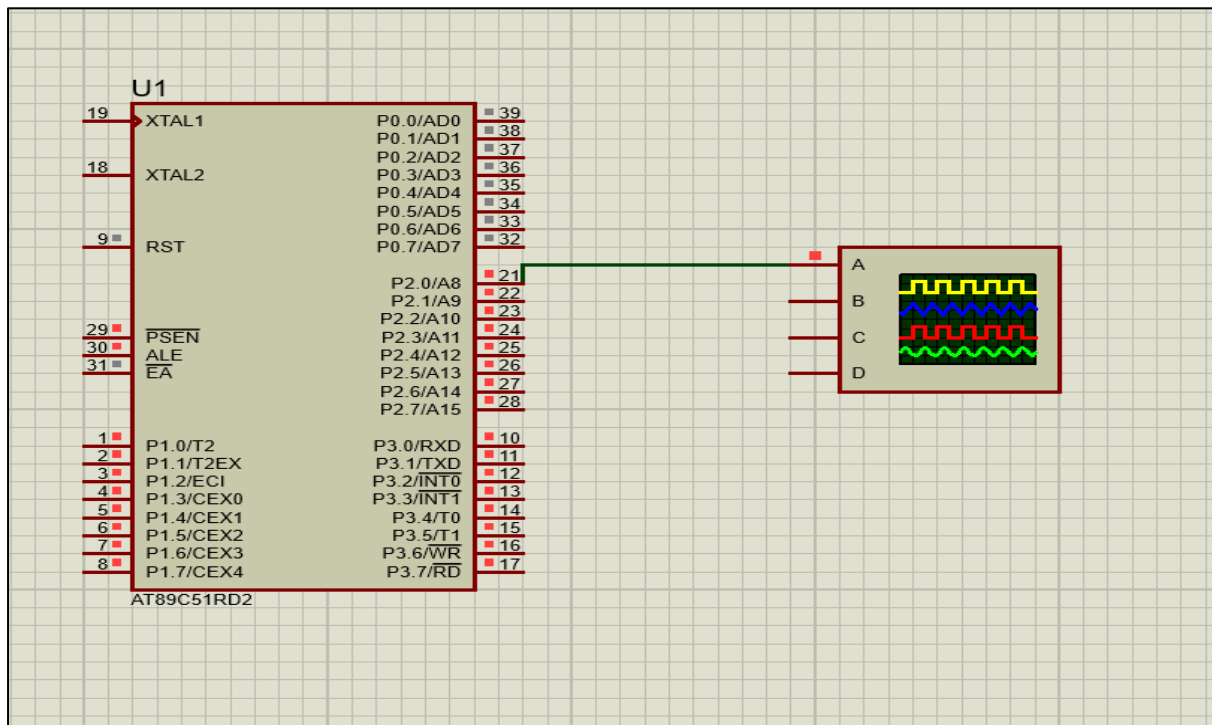
000A DELAY:
000A MOV R4,#35
000C H1:
000C MOV R3,#255
000E H2:
000E DJNZ R3,H2
0010 DJNZ R4,H1
0012 RET
-----END
```

OUTPUT:



Exp No: 2 GENERATION OF SQUARE WAVE USING PROTEUS

OUTPUT:



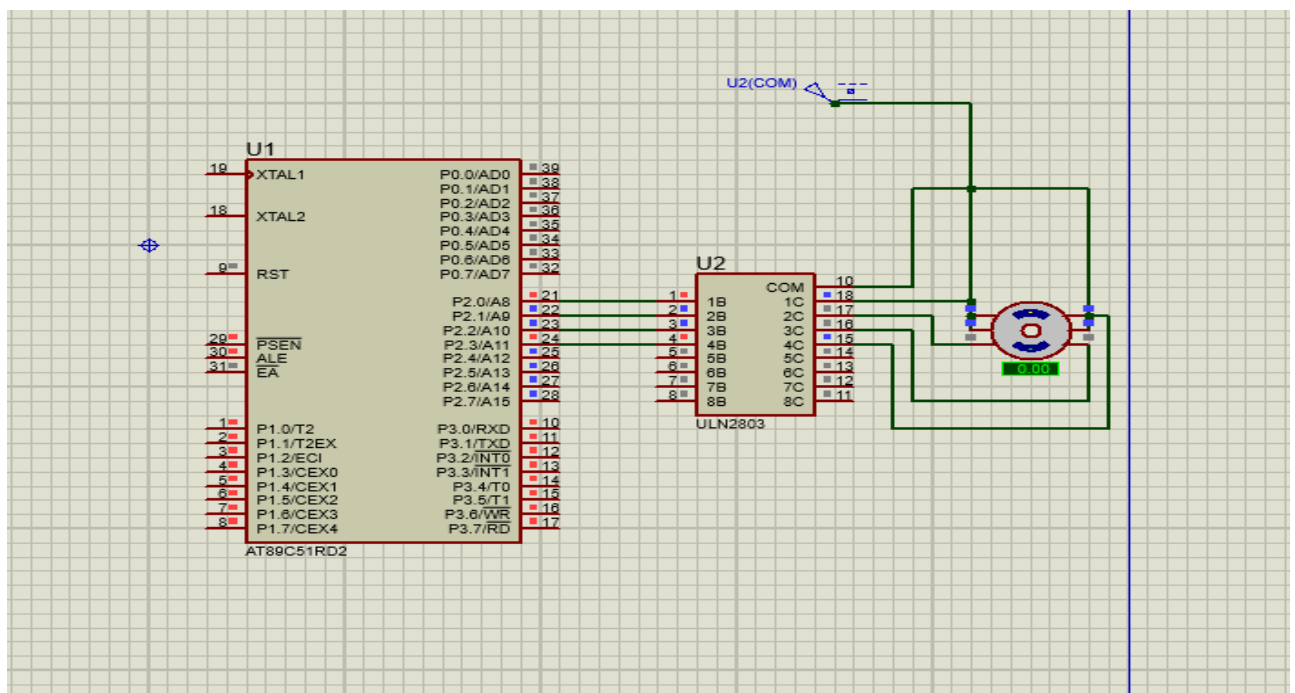
SOURCE CODE:

The screenshot shows the 'Source Code' window in Proteus 8 Professional. The title bar reads 'New Project.2 - Proteus 8 Professional - Source Code'. The menu bar includes 'File', 'Project', 'Build', 'Edit', 'Debug', 'System', and 'Help'. The toolbar contains various icons for file operations and debugging. The active tab is 'Source Code'. Below the tab, the window title is '8051 CPU Source Code - U1'. The file path is 'C:\Users\SAMRAK~1\AppData\Local\Temp\VSM Studio\9aa8303c576845669f57165c2992fef9\AT89C51RD2\Debug\Debug.SDI'. The assembly code is as follows:

```

----- ORG 0000H
0000 UP: SETB P2.0
0002 ACALL DELAY
0004 CLR P2.0
0006 ACALL DELAY
0008 SJMP UP
000A DELAY: MOV R4, #35
000C H1: MOV R3, #255
000E H2: DJNZ R3, H2
0010 DJNZ R4, H1
0012 RET
000E END
  
```

Exp No: 3 STEPPER MOTOR USING AT89C51 USING PROTEUS



The screenshot shows the Proteus 8 Professional Source Code window. The title bar reads "New Project.3 - Proteus 8 Professional - Source Code". The menu bar includes File, Project, Build, Edit, Debug, System, and Help. The toolbar contains various icons for file operations, debugging, and simulation. The main window has two tabs: "Schematic Capture" and "Source Code". The "Source Code" tab is active, displaying assembly code for an 8051 CPU. The code is organized into sections with labels like START, L1, L2, and RET. The address 0000H is marked as the origin. The code includes instructions such as MOV, ACALL, SJMP, DJNZ, and RET. The address 000E is marked as the end of the code.

8051 CPU Source Code - U1

C:\Users\SAMRAK~1\AppData\Local\Temp\VSM Studio\5497284508e34fc2805bd3720f85e09d\AT89C51RD2\Debug\Debug.SDI

```

----- ORG 0000H
0000      START:
0000      MOV P2,#09H
0003      ACALL DELAY
-----
0005      MOV P2,#0CH
0008      ACALL DELAY
-----
000A      MOV P2,#06H
000D      ACALL DELAY
-----
000F      MOV P2,#03H
0012      ACALL DELAY
-----
0014      SJMP START
-----
0016      DELAY:
0016      MOV R5,#30
-----
0018      L1:
0018      MOV R4,#255
-----
001A      L2:
001A      DJNZ R4,L2
001C      DJNZ R5,L1
-----
001E      RET
-----
000E      END

```

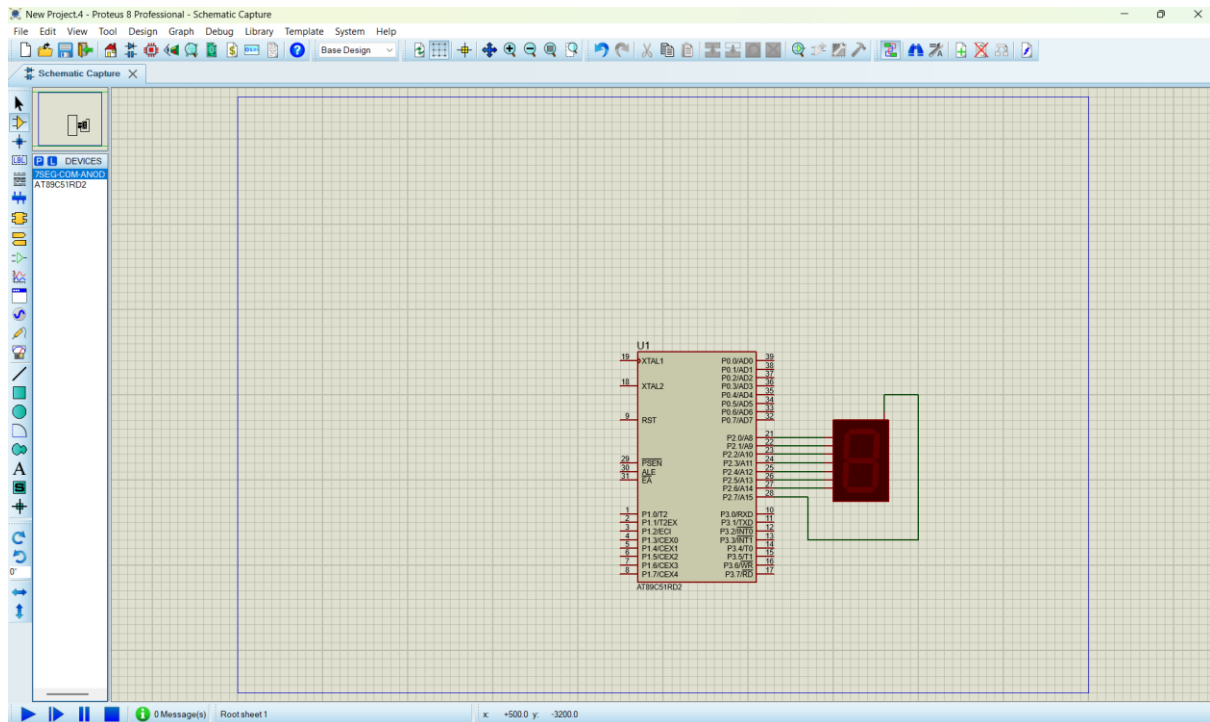
The diagram illustrates the connection of an AT89C51RD2 microcontroller to a 7-segment display via a ULN2803 driver. The microcontroller's pins are configured as follows:

- XTAL1** (pin 19) and **XTAL2** (pin 18) are connected to a crystal oscillator.
- RST** (pin 9) is connected to a reset circuit.
- PSEN** (pin 28), **ALE** (pin 30), and **EA** (pin 31) are connected to memory chips.
- P1.0/T2** (pin 1), **P1.1/T2EX** (pin 2), **P1.2/CE0** (pin 3), **P1.3/CE0** (pin 4), **P1.4/CEX1** (pin 5), **P1.5/CEX2** (pin 6), **P1.6/CEX3** (pin 7), and **P1.7/CEX4** (pin 8) are connected to various peripheral devices.
- P0.0/AD0** (pin 39) through **P0.7/AD7** (pin 42) are connected to a data bus.
- P2.0/A8** (pin 21), **P2.1/A9** (pin 22), **P2.2/A10** (pin 23), **P2.3/A11** (pin 24), **P2.4/A12** (pin 25), **P2.5/A13** (pin 26), **P2.6/A14** (pin 27), and **P2.7/A15** (pin 28) are connected to a data bus.
- P3.0/RXD** (pin 10), **P3.1/TXD** (pin 11), **P3.2/INT0** (pin 12), **P3.3/INT1** (pin 13), **P3.4/T0** (pin 14), **P3.5/T1** (pin 15), **P3.6/WR** (pin 16), and **P3.7/RD** (pin 17) are connected to various peripheral devices.

The ULN2803 driver (U2) is connected to the 7-segment display (U3) as follows:

- COM** (pin 10) is connected to the common anode of the display.
- 1B** (pin 18) is connected to the display's **a** segment.
- 2B** (pin 20) is connected to the display's **b** segment.
- 3B** (pin 22) is connected to the display's **c** segment.
- 4B** (pin 24) is connected to the display's **d** segment.
- 5B** (pin 26) is connected to the display's **e** segment.
- 6B** (pin 28) is connected to the display's **f** segment.
- 7B** (pin 30) is connected to the display's **g** segment.
- 8B** (pin 32) is connected to the display's **a** segment.

Exp No: 4 7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS



SOURCE CODE:

```
New Project.4 - Proteus 8 Professional - Source Code
File Project Build Edit Debug System Help
Schematic Capture Source Code
8051 CPU Source Code - U1
C:\Users\SAMRAK~1\AppData\Local\Temp\VSM Studio\e4cf3ae8d48c4aa581d6ba1167a5de15\AT89C51RD2\Debug\Debug.SDI

-----ORG 000H
0000 UP:
0000 MOV P2,#0C0H
0003 ACALL DELAY
-----
0005 MOV P2,#0F9H
0008 ACALL DELAY
-----
000A MOV P2,#0A4H
000D ACALL DELAY
-----
000F MOV P2,#0B0H
0012 ACALL DELAY
-----
0014 MOV P2,#99H
0017 ACALL DELAY
-----
0019 MOV P2,#92H
001C ACALL DELAY
-----
001E MOV P2,#82H
0021 ACALL DELAY
-----
0023 MOV P2,#0F8H
0026 ACALL DELAY
-----
0028 MOV P2,#80H
002B ACALL DELAY
-----
002D MOV P2,#90H
0030 ACALL DELAY
-----
0032 SJMP UP
-----
0034 DELAY:
0034 MOV R5,#10
-----
0036 H1:
0036 MOV R4,#180
-----
0038 H2:
0038 MOV R3,#255
-----
003A H3:
003A DJNZ R3,H3
003C DJNZ R4,H2
003E DJNZ R5,H1
-----
0040 RET
-----
000E END
```

OUTPUT:

